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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/283,371	09/21/2023	Theresa KRUSE	2023-1525A	6922

513 7590 07/21/2026

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EXAMINER	
TOTH, KAREN E	

ART UNIT	PAPER NUMBER
3791	

NOTIFICATION DATE	DELIVERY MODE
07/21/2026	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 18/283,371	Applicant(s) KRUSE et al.	
	Examiner KAREN E TOTH	Art Unit 3791	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 29 May 2026.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☑ This action is **FINAL.** 2b) ☐ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 19-38 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 19-38 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☐ The drawing(s) filed on _____ is/are: a)☐ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☐ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Claim Objections

2. The following is a quotation of 35 U.S.C. 112(d):

> (d) REFERENCE IN DEPENDENT FORMS.—Subject to subsection (e), a claim in dependent form shall contain a reference to a claim previously set forth and then specify a further limitation of the subject matter claimed. A claim in dependent form shall be construed to incorporate by reference all the limitations of the claim to which it refers.

The following is a quotation of pre-AIA 35 U.S.C. 112, fourth paragraph:

> Subject to the following paragraph [i.e., the fifth paragraph of pre-AIA 35 U.S.C. 112], a claim in dependent form shall contain a reference to a claim previously set forth and then specify a further limitation of the subject matter claimed. A claim in dependent form shall be construed to incorporate by reference all the limitations of the claim to which it refers.

3. Claims 19-38 are objected to because of the following informalities:

A series of singular dependent claims is permissible in which a dependent claim refers to a preceding claim which, in turn, refers to another preceding claim.

A claim which depends from a dependent claim should not be separated by any claim which does not also depend from said dependent claim. It should be kept in mind that a dependent claim may refer to any preceding independent claim. At least claims 35 and 38 are improperly presented in the claims. In general, applicant's sequence will not be changed. Future submissions which do not properly present claims may be treated as non-responsive. See MPEP § 608.01(n).

Appropriate correction is required.

Claim Rejections - 35 USC § 112

4. The following is a quotation of 35 U.S.C. 112(b):

(b) CONCLUSION.—The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the inventor or a joint inventor regards as the invention.

The following is a quotation of 35 U.S.C. 112 (pre-AIA), second paragraph:

The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the applicant regards as his invention.

5. Claims 19-38 are rejected under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph, as being indefinite for failing to particularly point out and distinctly claim the subject matter which the inventor or a joint inventor (or for applications subject to pre-AIA 35 U.S.C. 112, the applicant), regards as the invention.

Claim 19 calls for "applying at least two state transition models to assign a transport fluid glucose value in the transport fluid to the tissue glucose value"; it is unclear what "assigning" one value to another value might involve. Further, it is unclear what this transport fluid glucose value is or where or how it is obtained – is this a preset or constant value, or something that is measured? Does "assigning" the transport value to the tissue value using a state transition model somehow involve adjusting the tissue value? Further, if the "assigning" of the transport value to the tissue value changes the tissue value in some way, this renders the remainder of the claim unclear as the claim does not clearly indicate whether subsequent references to "the tissue glucose value" are referring to the value to which a transport value

has been assigned or to the originally determined tissue glucose value. The same issue is found in claim 37.

Claim 19 continues by calling for "estimating the current transport fluid glucose value based on an approximation of one of the at least two state transition models and the tissue glucose value", where the estimating uses one filter "if the measurement model has a linear function" and another filter "if the measurement model has a non-linear function". The measurement model (recited in the "determining a tissue glucose value" clause") is not recited as having any particular functions, let alone one of a linear or non-linear function.

Claim 19 continues by calling for "switching between the at least two state transition models depending on a rate of change over time of the current transport fluid glucose value". It is entirely unclear where or how a "rate of change" is involved in the method as claimed. Only one "current transport fluid glucose value" is recited as being estimated; as a whole, the method only calls for obtaining "at least two measured values" such that it is not readily apparent that even duplicating earlier steps in the method could result in sufficient "current transport fluid glucose values" to then enable determining a rate of change.

Still further, the claim calls for switching between the state transition models based on the rate of change, but the previous clause calls for selecting the state transition model based on a characteristic of the measurement model. It does not appear that there is any mechanism for the measurement model itself to change, such that it is entirely unclear how or why the claim calls for the contradictory step of switching between state transition models when the claim also requires the state transition model itself to be determined by the measurement model.

Still further, it is unclear how or why one would include "switching" between state transition models that are used to estimate "the current transport fluid glucose value" when only one of these values is found – it is unclear if the single tissue glucose value used as an input in this process is used multiple times to obtain a plurality of "current" values, or if the intent is to use determine and use multiple tissue glucose values using the measurement model(s?) with different measurement models and different tissue glucose values resulting in the use of different state transition models. The same issues are found in claim 37.

Claim 19 calls for both "estimating the current transport fluid glucose value" and also "generate the current transport fluid glucose value". It is unclear if the "estimating" is considered generating, or if "generat[ing]" involves some other procedure.

Claim 20 defines that at least one of the models is based on "a constant glucose concentration; a constant change in glucose concentration; or a weighted sum of previous glucose concentrations". It is entirely unclear what these parameters might be or how they are obtained. No part of the method involves measuring or otherwise finding a "glucose concentration". It is unclear if a "constant glucose concentration" is some present value or an ideally observed condition involving glucose concentrations over time which do not change; the same is true of the "constant change in glucose concentration". Still further, as no glucose concentrations appear to exist within in the method it is entirely unclear how one could calculate or obtain "previous glucose concentrations" and then find a weighted sum. The same issue is found in claim 34. Clarification is required.

Claim 21 calls for "adjusting the measurement noise value"; it is unclear how this could take place. It is unclear if this merely involves applying some factor to an observed noise value,

or to changing the measurement conditions that would result in an adjustment to the noise value. Further, claim 19 only calls for a single use of the measurement noise value, such that it is unclear if the intent is to involve further iterations of the method of claim 19 where that noise value is changed in later iterations, or "adjusting" this value prior to performing the method of claim 19. Clarification is required.

Claim 22 refers to "a random sample of measured values"; it is unclear if these are the "at least two measured values" as found as part of claim 19 or if these are some other "measured values", and, if other "measured values", how they relate to or differ from the "measured values" of claim 19. Clarification is required.

Claim 23 calls for "using a statistical test to check whether a null hypothesis is not rejected"; it is entirely unclear how this relates to any other part of the method, or what step is being checked using this statistical test. Clarification is required.

Claim 25 calls for "checking the measured values for outliers"; as only "at least two measured values" are obtained, it is unclear whether this would even allow the method to be performed.

Claim 26 further calls for comparing these measured values to "limit values"; it is unclear what these limit values are and how they are determined or otherwise selected.

Claim 27 further calls for "rejecting one of the measured values which is not determined to be one of the outliers as a measurement error if at least two of the measured values were previously rejected". As claim 19 only calls for obtaining "at least two measured values" the claim needs to clearly define how many total or additional measured values are obtained that would allow this to take place. Clarification is required.

Claim 31 recites that "a trend in transport fluid glucose concentration is classified" based on the measurement series. Claim 19 does not involve finding any value(s) of transport fluid glucose concentration; at best, claim 19 calls for determining a single transport fluid glucose value. It is entirely unclear what inputs might be involved in determining this "trend" of concentrations.

Where there is a great deal of confusion and uncertainty as to the proper interpretation of the limitations of a claim, it would not be proper to reject such a claim on the basis of prior art. As stated in In re Steele, 305 F.2d 859, 134 USPQ 292 (CCPA 1962), a rejection should not be based on considerable speculation about the meaning of terms employed in a claim or assumptions that must be made as to the scope of the claims. See MPEP 2173.06. The claims cannot be further treated on the merits.

Claim Rejections - 35 USC § 101

6. 35 U.S.C. 101 reads as follows:

> Whoever invents or discovers any new and useful process, machine, manufacture, or composition of matter, or any new and useful improvement thereof, may obtain a patent therefor, subject to the conditions and requirements of this title.

7. Claims 19- 38 are rejected under 35 U.S.C. 101 because the claimed invention is directed to a judicial exception (i.e., a law of nature, a natural phenomenon, or an abstract idea) without significantly more. Claims 19-38 do not include additional elements that integrate the exception into a practical application of the exception or that are sufficient to amount to significantly more than the judicial exception for the reasons provided below which are in line

with the 2014 Interim Guidance on Patent Subject Matter Eligibility (Federal Register, Vol. 79, No. 241, p 74618, December 16, 2014), the July 2015 Update on Subject Matter Eligibility (Federal Register, Vol. 80, No. 146, p. 45429, July 30, 2015), the May 2016 Subject Matter Eligibility Update (Federal Register, Vol. 81, No. 88, p. 27381, May 6, 2016), and the 2019 Revised Patent Subject Matter Eligibility Guidance (Federal Register, Vol. 84, No. 4, page 50, January 7, 2019).

The analysis of claim 19 is as follows:

Step 1: Claim 19 is drawn to a process, which is one of the four statutory categories.

Step 2A - Prong 1: Claim 19 is drawn to an abstract idea in the form of a process that, under its broadest reasonable interpretation, covers performance of the limitations in the mind but for the recitation of generic computer components. Additionally or alternatively, claim 19 is drawn to an abstract idea in the form of mathematical algorithms and/or formulas.

In particular, claim 19 recites the following limitations:

[A1]: determining a tissue glucose value using the series of measurements given, based on a measurement model, in which, by means of the measurement model, measurements of the sensor device are correlated to the tissue glucose levels while taking into account measurement noise,

[B1]: providing state transition models, in which, by means of using all provided state transition models, at least one glucose level in the transport fluid is correlated to the tissue glucose value that have been determined using the measurement model, and

[C1]: estimating the current glucose level based on a state transition model and the tissue glucose level that has been determined,

[D1]: in which at least one Kalman filter or at least one extended Kalman filter is used to provide the current glucose level;

[E1]: switching between provided state transition models based on some rate of change over time of a single determined value

These elements [A1]-[E1] of claim 19 are drawn to an abstract idea because they are processes that, under their broadest reasonable interpretation, are mere steps that are capable of being mentally performed with the aid of pen and paper. For example, a skilled artisan is capable of denoising tissue glucose measurements using a generic model in order to determine a tissue glucose level, convert the tissue glucose level to a glucose level of a transport fluid based on a generic model while taking into noise into account, and converting the transport fluid glucose level to a current glucose level based on a moving horizon estimation method . Additionally or alternatively, the elements [A1]-[E1] are drawn to an abstract idea because they are mathematical algorithms and/or formulas.

Step 2A - Prong Two: Claim 19 does not recite additional elements that integrate the judicial exception into a practical application. Claim 19 recites the following additional elements:

[A2]: measuring, using a probe, a measurement series including at least two measured values, that are spaced apart in time

[B2]: wherein the Kalman filter or the extended Kalman filter is executed on a processor coupled to the probe

The element [A2] does not integrate the exception into a practical application of the exception because the element amounts to merely adding insignificant extra-solution activity to the judicial exception, e.g., mere data gathering at a higher level of generality in conjunction with the abstract idea that uses conventional, routine, and well known elements - see MPEP 2106.04(d); MPEP 2106.05(g).

The element [B2] also does not integrate the exception into a practical application of the exception because the element amounts to merely providing a generic technological environment for execution of the abstract idea itself.

This extra-solution activity does not improve the functioning of technology, apply the judicial exception to effect a particular treatment, apply the judicial exception with a particular machine, or effect any sort of transformation.

Accordingly, the additional elements do not integrate the abstract into a practical application because they do not impose any meaningful limitations on practicing the abstract idea.

Step 2B: Claim 19 does not recite additional elements that amount to significantly more than the judicial exception itself. Claim 19 recites the following additional elements:

[A2]: determining, using a sensor device, a series of measurements comprising at least two measurements separated in time for a tissue glucose level in the tissue surrounding the transport fluid

[B2]: wherein the Kalman filter or the extended Kalman filter is executed on a processor coupled to the probe

The element [A2] does not qualify as significantly more because the element amounts to merely adding insignificant extra-solution activity to the judicial exception, e.g., mere data gathering at a higher level of generality in conjunction with the abstract idea that uses conventional, routine, and well known elements - see MPEP 2106.05(g). Additionally, the element is well-understood, routine, and conventional. US 5,408,999 A (Singh) teaches, in Col. 1, lines 41-57, fiber optic devices for measurement of blood gases, pH, electrolytes, and glucose are well known, and certain prior art sensors usually include an indicator molecule (dye) such as fluorescent or absorption dye which interacts with the component to be sensed or measured. The Examiner asserts that such fiber optic devices are capable of providing a series of measurements of tissue glucose levels.

The element [B2] does not qualify as significantly more as it merely recites well-understood, routine, and conventional processing elements used to provide a technological environment for execution of the abstract idea itself. See MPEP 2106.05(d), Bancorp Services v. Sun Life, 687 F.3d 1266, 1278, 103 USPQ2d 1425, 1433 (Fed. Cir. 2012) ("The computer required by some of Bancorp's claims is employed only for its most basic function, the performance of repetitive calculations, and as such does not impose meaningful limits on the scope of those claims."), showing that these computer functions are well-understood, routine, and conventional functions.

In view of the above, the additional elements individually do not amount to significantly more than the above-judicial exception (the abstract idea). Looking at the limitations as an ordered combination (that is, as a whole) adds nothing that is not already present when looking at the elements taking individually. There is no indication that the combination of elements

improves the functioning of a computer, for example, or improves any other technology. There is no indication that the combination of elements permits automation of specific tasks that previously could not be automated. There is no indication that the combination of elements includes a particular solution to a computer-based problem or a particular way to achieve a desired computer-based outcome. Rather, the collective functions of the claimed invention merely provide conventional computer implementation, i.e., the computer is simply a tool to perform the process.

Independent claim 37 recites mirrored device and storage medium limitations and is not patent eligible for substantially similar reasons.

Claims 20-36 and 38 depend from claim 19, and they recite the same abstract idea as claim 19. Furthermore, these claims only contain recitations that further limit the abstract idea (that is, the claims only recite limitations that further limit the mental process or mathematical algorithm) and/or append abstract ideas (that is, the claims only recite limitations that add further mental processes or mathematical algorithms). Claims 32 and 36 define the probe used for data gathering, still at a high level of generality and only for the insignificant extrasolution activity of data gathering. Claims 36 and 38 recite a generic technological environment (memory, computer-readable medium having instructions) only for the execution of the abstract idea, and do not add anything significantly more.

In view of the above, the additional elements do not integrate the abstract idea into a practical application and do not amount to significantly more than the above-judicial exception (the abstract idea). Looking at the limitations as an ordered combination (that is, as a whole) adds nothing that is not already present when looking at the elements taking

individually. There is no indication that the combination of elements improves the functioning of a computer, for example, or improves any other technology. There is no indication that the combination of elements permits automation of specific tasks that previously could not be automated. There is no indication that the combination of elements includes a particular solution to a computer-based problem or a particular way to achieve a desired computer-based outcome. Rather, the collective functions of the claimed invention merely provide conventional computer implementation, i.e., the computer is simply a tool to perform the process.

Response to Arguments

8. Applicant's arguments filed 29 May 2026 have been fully considered but they are not persuasive.

Applicant generally asserts that all objections and 112 rejections of the claims have been resolved by the claim amendments; as noted above, these amendments have only rendered the scope of the claims even more indefinite.

Regarding the rejections under 101, Applicant generally asserts that this has also been obviated by the amendments to the claims, but then also asserts that the use of either of a Kalman filter or an extended Kalman filter provides the practical application of a technological improvement to the general field of glucose monitoring. It should be noted that integration into a practical application must be provided by an additional element, not part of the abstract idea itself (see MPEP 2106.04(d)(2)), such that these remarks are moot.

Applicants remarks regarding the previously applied references are noted, but, as the scope of the claims is now too indefinite to allow examination on the merits, these remarks will

not be addressed at this time but will be revisited if still applicable once the other significant issues in the claims have been resolved.

Conclusion

9. No art has been applied against the claims at this time; however, as all claims are rejected under 112 and 101 above they are **not presently allowable** and the question of prior art will be revisited if applicable once the scope of the claims is resolved.

10. Applicant's amendment necessitated the new ground(s) of rejection presented in this Office action. Accordingly, **THIS ACTION IS MADE FINAL**. See MPEP § 706.07(a). Applicant is reminded of the extension of time policy as set forth in 37 CFR 1.136(a).

A shortened statutory period for reply to this final action is set to expire THREE MONTHS from the mailing date of this action. In the event a first reply is filed within TWO MONTHS of the mailing date of this final action and the advisory action is not mailed until after the end of the THREE-MONTH shortened statutory period, then the shortened statutory period will expire on the date the advisory action is mailed, and any nonprovisional extension fee (37 CFR 1.17(a)) pursuant to 37 CFR 1.136(a) will be calculated from the mailing date of the advisory action. In no event, however, will the statutory period for reply expire later than SIX MONTHS from the mailing date of this final action.

11. Any inquiry concerning this communication or earlier communications from the examiner should be directed to KAREN E TOTH whose telephone number is (571)272-6824. The examiner can normally be reached Mon - Fri 9a-6p.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Jennifer Robertson can be reached at 571-272-5001. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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/KAREN E TOTH/
Examiner, Art Unit 3791